Girls Build Bots — Junior Automotive Design

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Automotive Engineering — Automotive Design **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on craft/cardboard supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: from idea to road

The Automotive series follows a vehicle through the real industry pipeline: **design** it for a user and purpose, engineer it to perform, then manufacture it at scale. Juniors are ready for the real tension designers manage: style vs. function vs. safety.

Vocabulary: designer, user, function, style, safety, constraint, concept sketch, trade-off.

Automotive Design (90–120 minutes)

Badge theme: Designing a vehicle for a specific user and purpose — balancing style, function, and safety the way real automotive designers do.

Materials

- Design brief cards (a rural mail carrier; a family with a wheelchair user; a mobile dog groomer; a farmers-market vendor)
- Sketch sheets (side view + top view outlines), markers
- Cardboard, foam board, bottle caps, skewers, tape for concept models
- A “trade-off triangle” poster: STYLE / FUNCTION / SAFETY at the corners

Activity 1 — Every Design Is a Trade-Off (15 min)

Compare three real vehicle types (sports car, minivan, delivery truck) against the trade-off triangle: which corner did each design favor, and what did it give up? A sports car trades cargo space for style and speed; a delivery truck trades style for function. **Designers don’t pick everything — they choose deliberately for their user.**

Activity 2 — The Design Brief (15 min)

Teams of 2–3 draw a brief card and unpack it like professionals: who's the user, what must the vehicle do (**function**), what would hurt them if it failed (**safety**), and what would make them love it (**style**)? Teams write three **must-haves** and one **nice-to-have** — that's a real requirements list, with constraints.

Activity 3 — Sketch Two, Choose One (25 min)

Each team sketches **two different concepts** for their brief (side + top view, features labeled). Committing to your first idea is the rookie mistake — comparing two forces real design thinking. Teams pick one, and must say what the winning concept trades away.

Activity 4 — Build the Concept Model (35 min)

Build the chosen design in cardboard/foam — a **concept model** that shows every must-have feature (a ramp that folds, a market stall that opens, weatherproof mail access). It doesn't need to drive; it needs to *communicate the design*.

Activity 5 — Design Review (10 min) + Wrap (5 min)

Each team presents: user, must-haves, the trade-off they chose, and the feature they're proudest of. Audience asks one "what about…?" question per team — designers defending choices is the real job.

> **Robot moment (optional):** Share for 60 seconds how the team designs its robot each season under the same triangle — speed vs. capability vs. not-breaking — and what this year's robot traded away.

Completion Checklist — Automotive Design

By the end of the session, to earn the badge, each Scout can: - [ ] Explain how style, function, and safety trade off in vehicle design - [ ] Turn a user's needs into a must-have requirements list - [ ] Compare two concept sketches and justify a choice - [ ] Build and present a concept model that shows the design's key features

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Cardboard and sketches carry the session; the "robot moment" is a 60-second add-on.
- **Two sketches, always.** Enforce it kindly — it's the single best design habit in the session.
- **Push the trade-off question.** "What did your design give up?" separates designing from decorating.
- **Users with real constraints** (the wheelchair-accessible brief especially) produce the most thoughtful designs — give those briefs your strongest teams or extra support.

- **Model good design reviews.** Ask curious “what about…?” questions, never gotchas.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit “Girls Build Bots” anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the “Girls Build Bots” Girl Scout Gold Award project. Original curriculum aligned to the Junior Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.